



Producing Secure PDF Files

version 11.2

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Adding Security to Your PDF Files

Version 10.3
Feature 1317

You can make your PDF documents more secure by encrypting them and by adding security settings. Secure PDF documents may be required for electronic bills, confidential documents (like medical records), and other documents containing sensitive material (bank statements, loan applications, and so on).

Security settings let you assign passwords for opening and modifying the document and control access to printing, editing, and annotating the document.

A document can have both an open password and an owner password. You can view a document with either type of password, but you must enter the owner password to change the document or its security settings.

If you use the security settings to restrict access to certain features, all toolbar and menu items related to those features are dimmed in Adobe Reader or Acrobat.

To enforce these restrictions, the content of the document is encrypted using a 40- or 128-bit algorithm specified by Adobe.

This document provides information on the following:

- [Installation on page 2](#)
- [Using the PDFKey Tool on page 11](#)
- [Using the PDFKEYGEN Function on page 12](#)
- [Example Security Settings on page 13](#)
- [Tips on page 15](#)

INSTALLATION The PDF security features are built into the PDFLIB library so there are no files to install.

CONFIGURING THE INI FILES

In each PrtType:XXX control group for which you want security features, add this INI option:

```
< PrtType:PDF >
  Encrypt = Yes
```

This option tells the PDF library to load the PDFCRYPT library and encrypt PDF files. Encrypting the PDF file changes the file so it is no longer easy to read when transmitted over the network. It also means you cannot use a text editor to alter the file without destroying it.

Since, however, no passwords or permissions have been specified, this option provides only a minimal amount of security. If someone gets a copy of the file, there is no way to prevent that person from viewing the file in Acrobat Reader or altering it with the full Acrobat product.

You can, however, create more secure documents by including additional options in the PDF printer control group of your INI file.

In the same control group as the Encrypt option, you can add the SecurityGroup option to specify a control group which will contain the permissions, passwords, and encryption strength. Here is an example:

```
< PrtType:PDF >
  SecurityGroup = PDF_Encryption
  Encrypt = Yes
< PDF_Encryption >
  .
  .
```

NOTE While the only way to specify passwords is through INI options, the PDFKEYGEN built-in function lets you create a custom built-in that supplies the actual passwords. This way, the only thing that the INI file contains is

```
OwnerKey = ~PDFKEYGEN ~CUSTOMPASSWORDRULE
```

Keep in mind the INI files are generally inside a firewall and the passwords are stored in encrypted form.

Setting up a Security Control Group

The SecurityGroup option specifies the control group where permissions, passwords, and encryption strength are set. The permissions let you control if users can...

- Print the document. (You can also control the print quality.)
- Modify the document.
- Copy text or graphics to the clipboard.
- Add or update annotations.

- Fill in form fields.
- Access the document with accessibility tools, such as text to speech applications.
- Add navigational aids to the document.

This table shows the options you can set. All AllowXXX options default to Yes, meaning the permission is granted. See also [Understanding permissions on page 6](#)

Option	Description
AllowPrinting	Lets the user print the document.
AllowModify	Lets the user modify the document.
AllowCopy	Lets the user copy data to the clipboard.
AllowAnnotate	Lets the user add or update annotations.
AllowFormFields	Lets the user fill in form fields. You must set the KeyLength to 128 to use this option.
AllowAccessibility	Lets accessibility tools, such as text to speech applications, access the document. You must set the KeyLength to 128 to use this option.
AllowAssembly	Lets the user add navigational elements to the document. You must set the KeyLength to 128 to use this option.
AllowHighQualityPrinting	If you set AllowPrinting to Yes, set this option to Yes to let the user generate a high quality hard copy of the document. If you set AllowPrinting to Yes and this option to No, users can only print draft quality hard copy. You must set the KeyLength to 128 to use this option.
OwnerKey	Specifies the password required to change the document or its security settings. The password takes the form of an encrypted 64-byte hexadecimal encoded string. See Choosing passwords on page 4 for more information.
UserKey	Specifies the password required to open the document. The password takes the form of an encrypted 64-byte hexadecimal encoded string. See Choosing passwords on page 4 for more information.

Option	Description
KeyLength	Specifies the encryption strength, in bits, either 40 or 128. The default is 128. The key length must match the length provided to the PDFKey tool. The choice of key length is a primary factor in determining how secure the PDF file will be. Computers are fast enough that 40-bit encryption is susceptible to attacks in which every possible encryption key is tried. The 128-bit encryption is much more secure and allows finer control of permissions, but PDF files encrypted with a key length of 128 bits can only be viewed using Adobe Acrobat and Acrobat Reader 5.0 or later. Because Acrobat Reader is available at no cost, this restriction is generally no more than a minor inconvenience.

Choosing passwords

An encrypted version of the open and owner passwords as well as permission information is kept in the security control group. These take the form of 64-byte hexadecimal encoded strings. Here is an example:

```
6d6f62d768bc143cefa30fc4fd3cc00eb1f638157f1d985dd5fe8ebfaa7c8317
```

There are two ways to generate these keys:

- The PDFKey tool generates these keys (see [Using the PDFKey Tool on page 61](#) for more information). Because the permissions are encoded in the keys, the permissions in the Security control group *must* match those provided to the PDFKey tool.
- You can also use the PDFKEYGEN built-in function to generate the OwnerKey and UserKey information. See [Using the PDFKEYGEN Function on page 62](#) for more information.

You can set passwords any way you like. Choosing the password is up to you. This table describes the levels of security you get based on the passwords you set up:

Add this password

Owner	Open	To provide this level of security
Yes	Yes	You must enter the owner or open password to view the document. Users are bound by the assigned permissions You cannot change the security settings without the full Acrobat product and the owner password.
Yes	No	No password is required to view the document. Users are bound by the assigned permissions. You cannot change the security settings without the full Acrobat product and the owner password.
No	Yes	You must enter the open password to view the document Anyone with the full Acrobat product and the open password can view and change the PDF file, including the file's security settings.

Add this password

Owner	Open	To provide this level of security
-------	------	-----------------------------------

No	No	No password is required to view the document. Users are bound by the assigned permissions. Anyone with the full Acrobat product can change the security settings — so it makes little sense to set passwords in this manner.
----	----	--

NOTE If you forget a password, there is no way to retrieve it from the document. Be sure to store passwords in a secure location in case you forget them.

Understanding permissions

This table shows how the permissions are related.

If this is set to True Attempts to set the following to False are ignored

AllowModify	AllowAnnotate, AllowFormFields, and AllowAssembly
AllowAnnotate	AllowFormFields and AllowAssembly
AllowFormFields	AllowAssembly

This table provides a general overview of how permissions are related.

If this is set to True Attempts to set the following to False are ignored

AllowModify	AllowAnnotate, AllowFormFields, and AllowAssembly
AllowAnnotate	AllowFormFields and AllowAssembly
AllowCopy	AllowAccessibility
AllowFormFields	AllowAssembly

NOTE PDF files with security settings are called **secure** files. You cannot insert a secure PDF file into another PDF file. You can, however, insert a nonsecure PDF file into a secure PDF file. In this case, the nonsecure PDF file inherits the secure PDF file's security settings.

These tables show all permitted combinations of permissions for Adobe PDF files. If you set permissions differently than those shown below, Adobe changes them based on the information shown in these tables.

KeyLength	Modify	Annotations	FormFields	Assembly
40	Allowed	Allowed	Allowed	Allowed
40	Not allowed	Allowed	Allowed	Allowed
40	Not allowed	Not allowed	Allowed	Allowed
40	Not allowed	Not allowed	Not allowed	Allowed
128	Not allowed	Not allowed	Not allowed	Not allowed
128	Allowed	Allowed	Allowed	Allowed
128	Not allowed	Allowed	Allowed	Allowed
128	Not allowed	Not allowed	Allowed	Allowed
128	Not allowed	Not allowed	Not allowed	Allowed

KeyLength Modify Annotations FormFields Assembly

128	Not allowed	Not allowed	Not allowed	Not allowed
-----	-------------	-------------	-------------	-------------

KeyLength Copy Read Access

40	Allowed	Allowed
40	Not allowed	Allowed
40	Not allowed	Not allowed
128	Allowed	Allowed
128	Not allowed	Allowed
128	Not allowed	Not allowed

KeyLength Print

40	High resolution
40	None
128	High resolution
128	Low resolution
128	None

While testing the security feature of the PDF Converter, Skywire Software has noted an issue with Adobe's products in which permissions set correctly in the PDF Converter are displayed incorrectly on the Adobe security settings window. It appears the permissions work as expected, regardless of whether those settings display correctly.

There are, however, a couple of caveats that do not follow these permitted combinations:

For a 40-bit encryption

Adobe lets you set Annotation ~~to~~ *Not allowed* even though Modify was set ~~to~~ *Allowed*. Below are the permission settings for this caveat.

```

Encryption      = Yes
KeyLength        = 40
Print            = High resolution
Modify           = Allowed
Annotation       = Not allowed
Form Fields     = Allowed
Assembly        = Allowed
Copy            = Not allowed
Accessibility    = Not allowed

```

In addition...

- When you set AllowModify to *Allowed*, AllowAnnotations remains *Not allowed*, although it should be *Allowed*.
- When you set AllowAnnotations to *Allowed*, AllowAssembly remains *Not allowed*, although it should be *Allowed*.
- When you set AllowFormFields to *Allowed*, AllowFormFields and AllowAssembly both remain *Not allowed*, although they should be *Allowed*.
- When you set AllowAccessibility to *Allowed*, it remains *Not allowed*.
- When you set AllowAssembly to *Allowed*, it remains *Not allowed*.

For a 128-bit encryption

Adobe lets you set Modify to *Allowed*, even though Annotation and Form Fields were set to *Not allowed*. Below are the permission settings for this caveat.

```
Encryption      = Yes
KeyLength       = 128
Print           = Low resolution
Modify          = Allowed
Annotation      = Not allowed
Form Fields     = Not allowed
Assembly        = Allowed
Copy            = Not allowed
Accessibility   = Not allowed
```

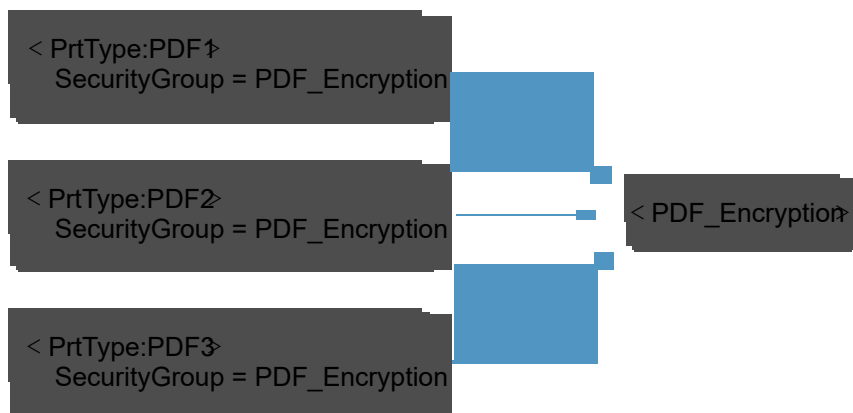
In addition...

- When you set AllowModify to *Allowed*, AllowAnnotations and AllowFormFields both remain *Not allowed*, although they be *Allowed*.
- When you set AllowCopy to *Allowed*, AllowAccessibility remains *Not allowed*.
- When you set AllowAnnotations to *Allowed*, AllowAssembly remains *Not allowed*, although it should be *Allowed*.
- When you set AllowFormFields to *Allowed*, AllowAssembly remains *Not allowed*, although it should be *Allowed*.

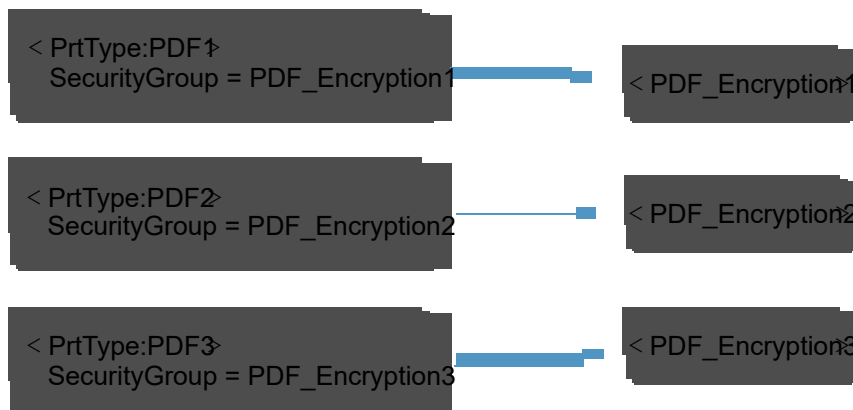
NOTE If you encounter any other unusual combinations, you may want to contact Adobe.

Setting up multiple security groups

You can set up your PDF PrtType:XXX control groups to share a common set of security settings or have security settings tailored to each control group. This illustration shows how you would set your printer options to use the same security settings:



Or, you can have custom security settings for each printer control group:



Choose the approach that works best in your situation.

Setting up document-level security

You can implement unique passwords at the document level. All you have to do is provide a custom `~PDFPASSWORD` function (the name can vary).

For each recipient in each transaction in each batch, the GenPrint program calls the PDFPrint function in the PDFW32.DLL file. The PDFPrint function checks the Encrypt option in the PrtType:PDF control group to see if encryption is desired.

If the Encrypt option is set to Yes, the PDFPrint function calls the PDFCryptInit function to an encryption *context*. If no passwords are passed to the PDFCryptInit function, as is the case with the PDF Converter, it checks the OwnerKey option:

```
< PDF_Encryption >  
    OwnerKey =
```

Keep in mind that the SecurityGroup control group must be specified in this option:

```
< PrtType:PDF>  
    SecurityGroup = PDF_Encryption
```

If the OwnerKey option is set to `~PDFKEYGEN ~PDFPASSWORD`, the INI built-in function calls the `~PDFPASSWORD` function, then passes the result to the `~PDFKEYGEN` function.

NOTE The `~PDFPASSWORD` function varies for each implementation and is usually part of CUSLIB.

USING THE PDFKEY TOOL

Use the PDFKey tool to generate the encrypted passwords used in the security control group.

Program names

Windows 32-bit PDFKW32.EXE

Place the executable file (PDFKW32.EXE) in the directory which contains the other Skywire Software binary files.

Syntax

pdfkw32 /U /O /K /P /F /M /C /N /R /A /? /H

Parameter Description

/U	Enter the password required to open the document. You can enter up to 32 characters. Passwords are case sensitive.
/O	Enter the password required to modify the document or its security settings. You can enter up to 32 characters. Passwords are case sensitive.
/K	Enter either 40 or 128 to specify the length of the encryption key. The default is 128.
/P	Enter No to prevent users from printing the file, L to permit low quality printing, or H to permit high quality printing.
/F	Enter No to prevent users from entering form fields. The default is Yes.
/M	Enter No to prevent users from modifying the document. The default is Yes.
/C	Enter No to prevent users from copying text from the document to the clipboard. The default is Yes.
/N	Enter No to prevent users from annotating the document. The default is Yes.
/R	Enter No to prevent users from using reader accessibility tools to view the document. The default is Yes.
/A	Enter No to prevent users from adding navigation aids, such as bookmarks. The default is Yes.
/? or /H	Enter either of these parameters to display information about the syntax and parameters you can use with this utility.

All parameters are case insensitive and can be preceded with either a backslash (\) or a dash (-). Omit spaces between an argument and its value. Passwords can, however, contain spaces. Simply enclose the entire password parameter in quotation marks, as shown here:

```
pdfkw32 "/O=Password With Spaces"
```

For more information about passwords see [choosing passwords on page.4](#)

USING THE PDFKEYGEN FUNCTION

You can use the PDFKEYGEN built-in function to generate the OwnerKey and UserKey information. When you include a string in this form...

```
OWNERPASS/USERPASS
```

this built-in function generates the OwnerKey and the UserKey values. The OwnerKey is returned via the function call. The UserKey is added to the INI context. To omit one or both of the passwords, modify the string as shown here:

```
"/USERPASS"  
"OWNERPASS/"  
"/"
```

To use the PDFKEYGEN built-in, first create a custom built-in that returns the password string in the format described above.

To call PDFKEYGEN (assuming the custom built-in function is registered as PDFPASSWORD), specify the OwnerKey INI option as shown here:

```
OwnerKey = ~PDFKEYGEN ~PDFPASSWORD
```

You can omit the UserKey option since it will be supplied via the above call.

NOTE You can find information on writing and registering built-in functions in the API documentation at:

<https://pd.docucorp.com/support/doc/rel102/api/coralib/INISys.htm#INIRegisterFunction>

EXAMPLE SECURITY SETTINGS

Here are some examples of how you can set up your PDF security configuration.

The examples include choosing permissions, passwords, and encryption strengths; generating keys with the PDFKey tool; and setting the appropriate INI options.

Example 1 This example shows how to set up:

- 128-bit encryption security
- An owner password of *Skywire*
- No password required to open the document
- Permissions that prevent readers from modifying information or copying information to the clipboard

Here is how you would run the PDFKey tool:

```
PDFKW32 /O=Skywire /K=128 /M=N /C=N
```

The output of the PDFKey tool is:

```
OwnerKey:
e551db056412e608eeab3e5f96619ac9296b49e419467fdb8879f75f95c2a816
UserKey:
605e0bb040c9ce39d650e718b3127f9b10000000fc0000000000000000000000
```

Here are the additions you would make to your INI file, assuming that the printer type control group is called PrtType:PDF.

```
< PrtType:PDF >
  Encrypt = Y
  SecurityGroup = PDF_Encryption_Example_1
< PDF_Encryption_Example_1 >
  KeyLength = 128
  OwnerKey =
e551db056412e608eeab3e5f96619ac9296b49e419467fdb8879f75f95c2a816
  UserKey =
605e0bb040c9ce39d650e718b3127f9b10000000fc0000000000000000000000
  AllowModify = No
  AllowCopy = No
```

You do not have to enter a password to view documents generated with these INI options. To modify a document, however, or to copy information from it to the clipboard, you must enter the owner password. Also, because 128-bit encryption option is used, you must have Acrobat 5.0 or later to view these documents.

Example 2 This example shows how to set up:

- 40-bit encryption security
- An owner password of *Skywire*
- A open password of *EncryptionIsFun*
- No permission restrictions

Here is how you would run the PDFKey tool:

```
pdfkw32 /o=Skywire /u=EncryptionIsFun /k=40
```

The output of the PDFKey tool is:

```
OwnerKey:  
90ed4c70598767459de9523a9ce7e77ac51f4459257401fbae1936b6b1bbc7fd  
UserKey:  
f6ac1d4b26000000000000000000000000200000005800000000000000000000
```

Here are the additions you would make to your INI file:

```
< PrtType:PDF >  
    Encrypt = Yes  
SecurityGroup = PDF_Encryption_Example_2  
< PDF_Encryption_Example_2 >  
    KeyLength = 40  
    OwnerKey =  
90ed4c70598767459de9523a9ce7e77ac51f4459257401fbae1936b6b1bbc7fd  
    UserKey =  
f6ac1d4b260000000000000000000000002000000058000000000000000000000
```

To view documents generated with these settings, you must enter the password *EncryptionIsFun* when prompted by Acrobat. Once the document opens, there are no restrictions.

To change the security settings, you must enter the owner password. **Skylwire**. Because 40-bit encryption is used, you only need Acrobat 4.0 or later to view documents created with these settings.

Example 3 This example shows how to set up:

- 128-bit encryption security
- No owner password
- A open password of *AnythingGoes*
- No permission restrictions

Here is how you would run the PDFKey tool:

PDFKW32 /u=AnythingGoes

Since the key length defaults to 128, the keys are:

```
OwnerKey:  
87958ea2053c6e89302ba926dfd78a2b3d0213d8e9569734d3045a8be297370e  
UserKey:  
75a73844c09078ad1a587957d4328e341000000083000000000000000000000
```

Here are the additions you would make to your INI file:

```
< PrtType:PDF >  
    Encrypt = Yes  
SecurityGroup = PDF_Encryption_Example_3  
< PDF_Encryption_Example_3 >  
    KeyLength = 128  
    OwnerKey =  
87958ea2053c6e89302ba926dfd78a2b3d0213d8e9569734d3045a8be297370e  
    UserKey =  
75a73844c09078ad1a587957d4328e341000000083000000000000000000000
```

To view documents created with these settings, you must enter the password *AnythingGoes* when prompted. You then have unrestricted use of the document. Since there is no owner password, you can even modify security options. Because you are using 128-bit encryption, you must have Acrobat 5.0 or later to open the document.

TIPS In case you run into problems, keep in mind...

- If you cannot open a password-protected PDF file after supplying the *correct* password, this probably indicates you have errors in your setup for producing secure PDF files.

The command line parameters used with the PDFKey tool, which produces the OwnerKey and UserKey hex strings, must match the secure PDF INI settings used when the PDF file is produced. Check your secure PDF INI settings for problems such as misspellings and INI settings that do not match the parameters used when running PDFKey. For example, specifying a 40-bit key length to PDFKey (/K=40) and using a KeyLength=128 INI setting will produce a PDF file that will not open in Acrobat.

- If you set the Print option (/P) to No in the PDFKey tool, you cannot set the AllowHighQualityPrinting option to No in the INI file. If you do, the result is an error, such as the one described above.
- When generating the hex key using the PDFKey tool, the system prints the output in INI format, as shown here:

```
D:\rel103>pdfkw32
< PDF_Encryption >
KeyLength = 128
OwnerKey =
36451bd39d753b7c1d10922c28e6665aa4f3353fb0348b536893e3b1db5c579b
UserKey =
7880927481fd184b32c0efb547ef5adb100000006c000000000000000000000000000000
```

The output is formatted this way so you can copy and paste these settings into your INI file.

- When configuring PDF encryption for use with IDS, you must modify two INI files: DAP.INI and the INI file for the MRL. The DAP.INI file contains all of the SecurityGroup names and SecurityGroups for all of the MRLs for which encryption is needed. The MRL's INI file contains only the SecurityGroup name and settings used by that MRL.

You can also set up multiple SecurityGroups within one MRL. Both the DAP.INI and MRL's INI file would contain the names of all SecurityGroups in the PrtType:PDF control group as well as the individual SecurityGroups.

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